Task 1. Data Immersion and Wrangling

step 1: Data Access and Familiarisation

1.1 Load the Dataset

In [9]:
```
import  numpy as np
import pandas as pd
import matplotlib.pyplot as  plt
import seaborn as sns
```

In [15]:
```
df = pd.read_csv('netflix_titles.csv')
df.head()
```

Out[15]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021

1.2 Check the structure

In [16]:

```
df.shape
df.info()
df.head(5)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Out[16]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021

step 2: Data Quality Assessment

2.1 check missing values

In [21]:
```
df.isnull().sum()
```

Out[21]:
```
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
rating             4
duration           3
listed_in          0
description        0
dtype: int64
```

2.2 Check duplicates and replace it

In [22]:
```
def clean_data(df):

    df = df.drop_duplicates()
    df['date_added'] = pd.to_datetime(df['date_added'],errors = 'coerce')
    df['country'] = df['country'].fillna('Unknown')
    df['director'] = df['director'].fillna('Unknown')

    return df
```

2.3 Dataset Size Before and After Cleaning

In [23]:
```
rows_before = df.shape[0]

df_clean = clean_data(df)

rows_after = df_clean.shape[0]

print("Rows before cleaning:", rows_before)
print("Rows after cleaning:", rows_after)
print("Duplicates removed:", rows_before - rows_after)
```

```
Rows before cleaning: 8807
Rows after cleaning: 8807
Duplicates removed: 0
```

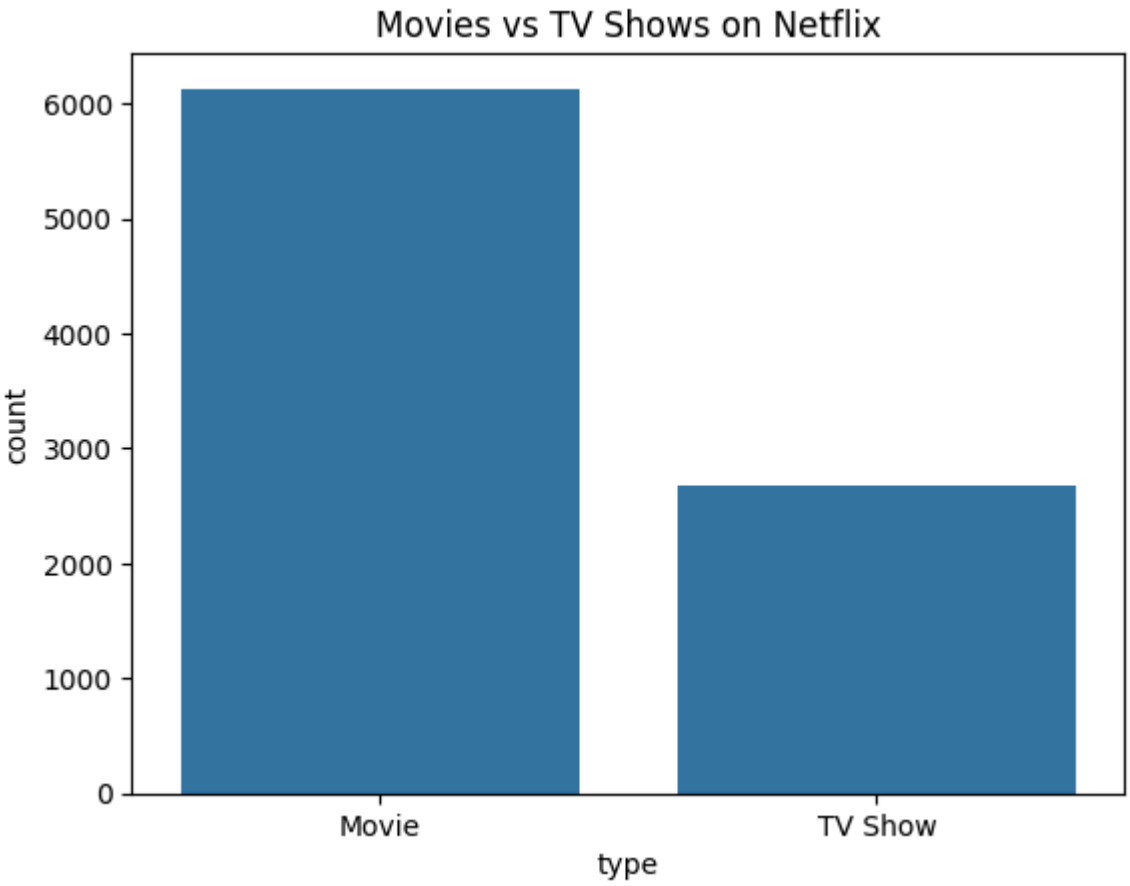
In [24]:
```
df['type'].value_counts()
```

Out[24]:
```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```

Distribution of Movies vs TV Shows on Netflix

In [25]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='type', data=df)
plt.title("Movies vs TV Shows on Netflix")
plt.show()
```

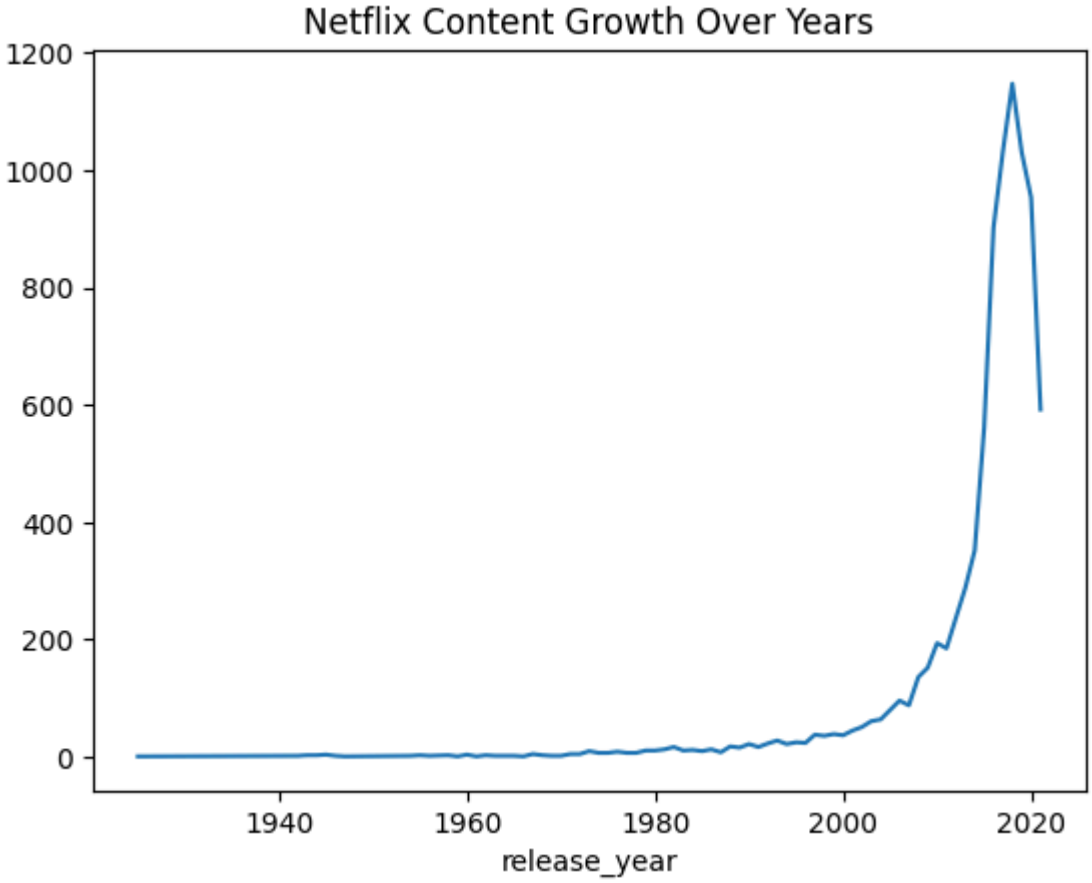


In [26]:
```
df['release_year'].value_counts().sort_index()
```

Out[26]:
```
release_year
1925       1
1942       2
1943       3
1944       3
1945       4
        ...
2017    1032
2018    1147
2019    1030
2020     953
2021     592
Name: count, Length: 74, dtype: int64
```

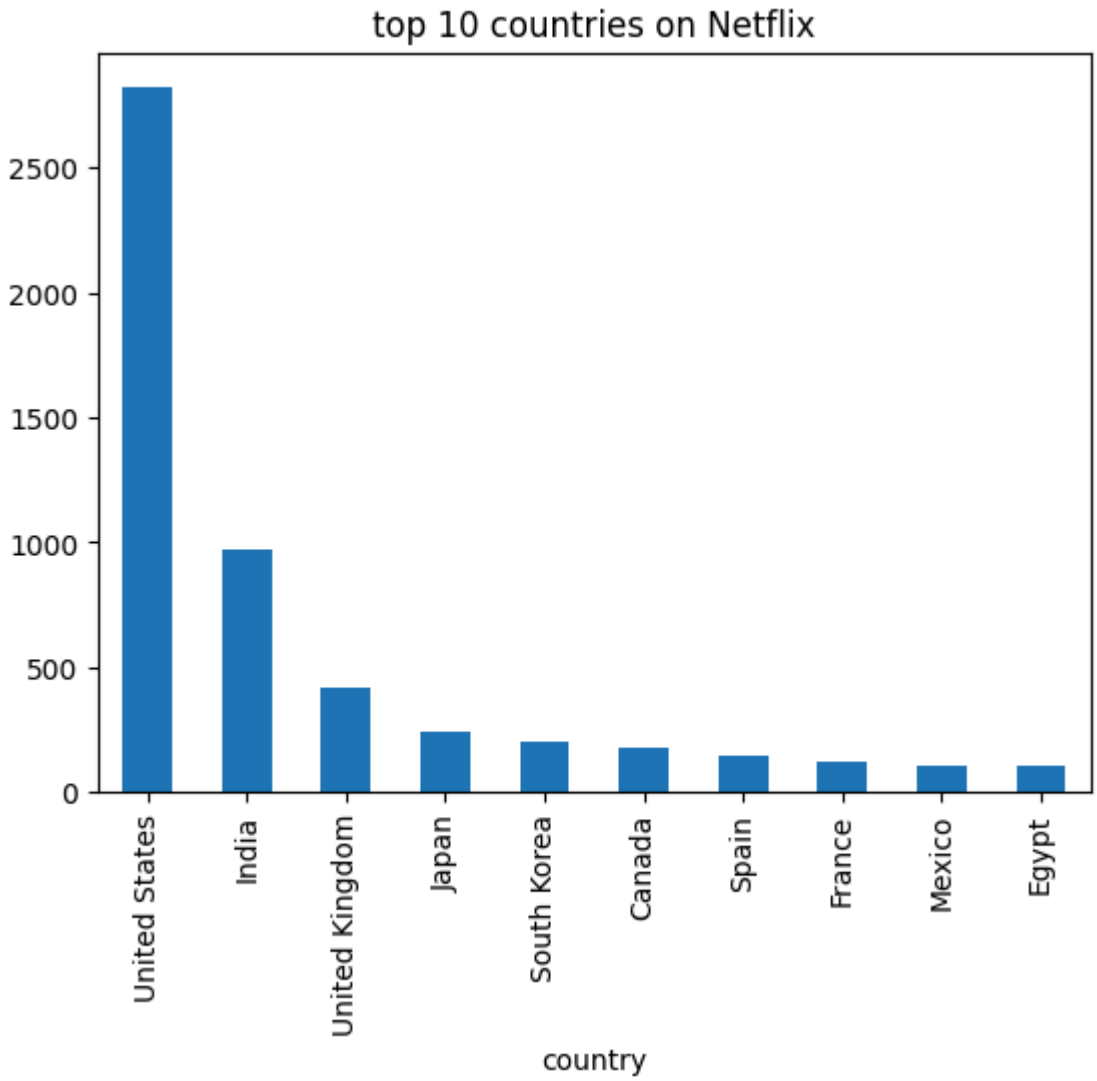
Netflix Content Growth Over the Years

In [27]:
```
df['release_year'].value_counts().sort_index().plot()
plt.title('Netflix Content Growth Over Years')
plt.show()
```



Top 10 Countries Producing Netflix Content

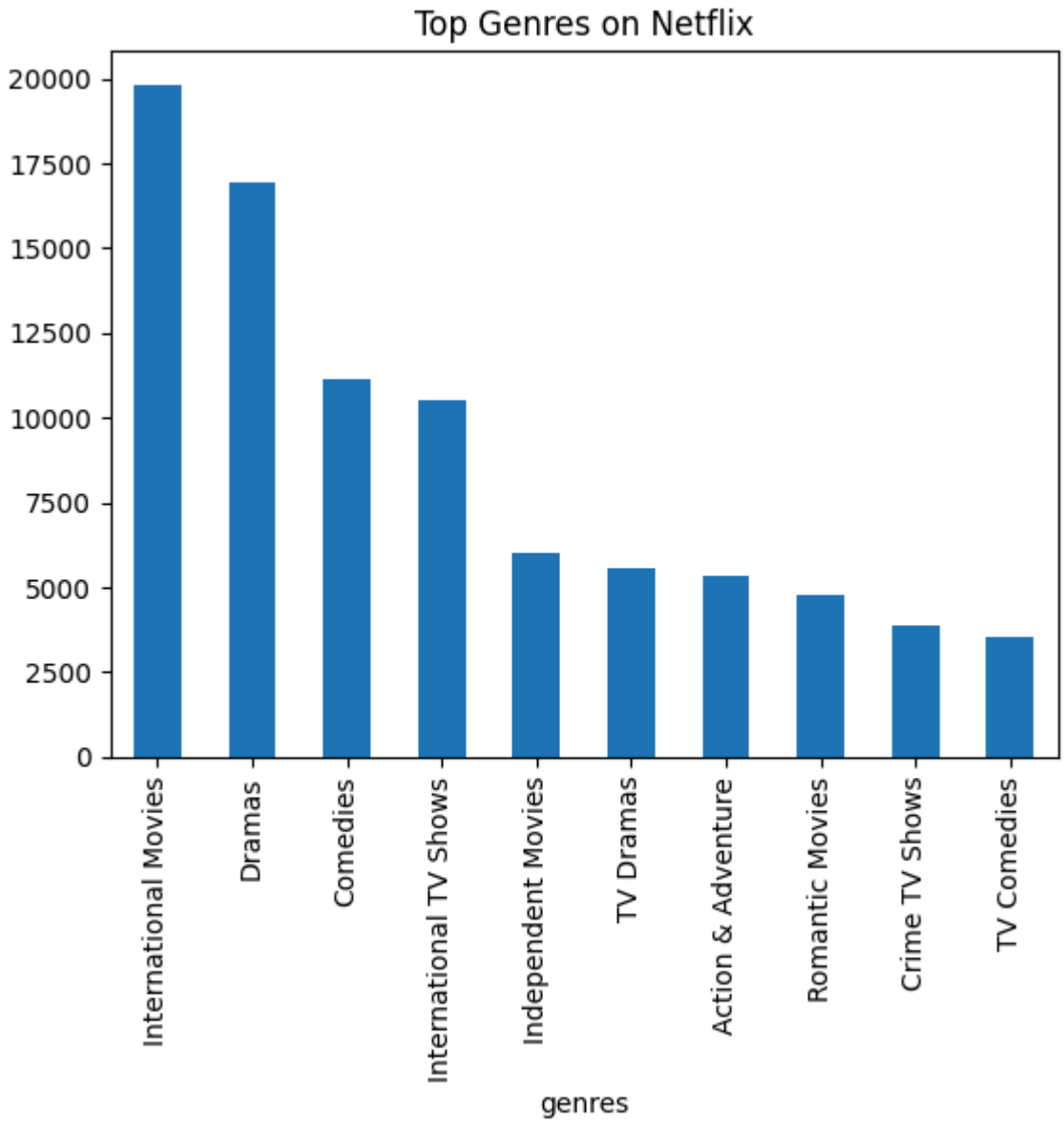
In [28]:
```
df['country'].value_counts().head(10).plot(kind = 'bar')
plt.title('top 10 countries on Netflix')
plt.show()
```



Top Genres on Netflix

In [34]:
```
df['genres'] = df['listed_in'].str.split(', ')
df = df.explode('genres')

df['genres'].value_counts().head(10).plot(kind='bar')
plt.title('Top Genres on Netflix')
plt.show()
```



Content Recommendation Based on Genre

In [36]:
```
def recommend(title,df):
    genre = df[df['title']== title]['listed_in'].values[0]
    recommendations = df[df['listed_in']==genre]['title'].head(5)
    return recommendations
```

In [37]:
```
recommend("Narcos", df)
```

Out[37]:
```
2415    Queen of the South
2415    Queen of the South
2415    Queen of the South
2415    Queen of the South
2415    Queen of the South
Name: title, dtype: object
```

In [38]:
```
df
```

Out[38]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015

	show_id	type	title	director	cast	country	date_added	release_year
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8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015

127167 rows × 13 columns

In [40]:
```
df.size
```

Out[40]:
```
1653171
```

Export Cleaned Dataset to CSV File

In [ ]:
```
df.to_csv('final_netflix.csv', index=False)
```